Topic: Substitutions

Question: Use a substitution to solve the separable differential equation.

$$y' = x + y$$

Answer choices:

A $y = Ce^x + x + 2$

B $y = Ce^x - x - 2$

C $y = Ce^x + x + 1$

D $y = Ce^x - x - 1$

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Solution: D

If we choose the substitution $u = ax + by$, then we can set up the substitution

$$u = x + y$$

$$u' = 1 + y'$$

and then solve this second equation $u' = 1 + y'$ for y' to get $y' = u' - 1$. Then we'll substitute $y' = u' - 1$ into the left side of the original differential equation, and $u = x + y$ into the right side of the original differential equation, and we'll get

$$u' - 1 = u$$

$$u' = u + 1$$

$$\frac{du}{dx} = u + 1$$

Separate variables,

$$du = (u + 1)\, dx$$

$$\frac{1}{u + 1}\, du = dx$$

then integrate both sides.

$$\int \frac{1}{u + 1}\, du = \int dx$$

$$\ln|u + 1| = x + C$$

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Now we'll solve for u.

$$e^{\ln|u+1|} = e^{x+C}$$

$$|u+1| = e^{x+C}$$

$$u+1 = Ce^{x}$$

$$u = Ce^{x} - 1$$

Back-substitute for $u = x + y$, then solve for y.

$$x + y = Ce^{x} - 1$$

$$y = Ce^{x} - x - 1$$

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Topic: Substitutions

Question: Use a substitution to solve the separable differential equation.

$$y' = 3x + y$$

Answer choices:

A $y = Ce^x + 3x + 3$

B $y = Ce^x - 3x - 3$

C $y = Ce^x + x + 1$

D $y = Ce^x - x - 1$

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Solution: B

If we choose the substitution $u = ax + by$, then we can set up the substitution

$$u = 3x + y$$

$$u' = 3 + y'$$

and then solve this second equation $u' = 3 + y'$ for y' to get $y' = u' - 3$. Then we'll substitute $y' = u' - 3$ into the left side of the original differential equation, and $u = 3x + y$ into the right side of the original differential equation, and we'll get

$$u' - 3 = u$$

$$u' = u + 3$$

$$\frac{du}{dx} = u + 3$$

Separate variables,

$$du = (u + 3)\, dx$$

$$\frac{1}{u + 3}\, du = dx$$

then integrate both sides.

$$\int \frac{1}{u + 3}\, du = \int dx$$

$$\ln|u + 3| = x + C$$

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Now we'll solve for u.

$$e^{\ln|u+3|} = e^{x+C}$$

$$|u+3| = e^{x+C}$$

$$u+3 = Ce^{x}$$

$$u = Ce^{x} - 3$$

Back-substitute for $u = 3x + y$, then solve for y.

$$3x + y = Ce^{x} - 3$$

$$y = Ce^{x} - 3x - 3$$

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Topic: Substitutions

Question: Use a substitution to solve the separable differential equation.

$$y' = 2x + 2y$$

Answer choices:

A $y = Ce^x - 2x - 2$

B $y = Ce^x - x - 1$

C $y = Ce^{2x} - x - \frac{1}{2}$

D $y = \frac{1}{2}Ce^x - x - \frac{1}{4}$

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Solution: C

If we choose the substitution $u = ax + by$, then we can set up the substitution

$$u = 2x + 2y$$

$$u' = 2 + 2y'$$

and then solve this second equation $u' = 2 + 2y'$ for y' to get $y' = (1/2)u' - 1$. Then we'll substitute $y' = (1/2)u' - 1$ into the left side of the original differential equation, and $u = 2x + 2y$ into the right side of the original differential equation, and we'll get

$$\frac{1}{2}u' - 1 = u$$

$$u' - 2 = 2u$$

$$u' = 2u + 2$$

$$\frac{du}{dx} = 2u + 2$$

Separate variables,

$$du = (2u + 2)\, dx$$

$$\frac{1}{2u + 2}\, du = dx$$

then integrate both sides.

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$$\int \frac{1}{2u+2}\,du = \int dx$$

$$\frac{1}{2}\ln|u+1| = x + C$$

Now we'll solve for u,

$$\ln|u+1| = 2x + C$$

$$e^{\ln|u+1|} = e^{2x+C}$$

$$|u+1| = e^{2x+C}$$

$$u + 1 = Ce^{2x}$$

$$u = Ce^{2x} - 1$$

Back-substitute for $u = 2x + 2y$, then solve for y.

$$2x + 2y = Ce^{2x} - 1$$

$$2y = Ce^{2x} - 2x - 1$$

$$y = Ce^{2x} - x - \frac{1}{2}$$

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